

A transition with renters and owners

Supporting proof for the illustrative model

1 Result and scope

There is an open set of economies with endogenous ownership, substantial renting, positive child goods costs, and a positive rebated property tax in which a small permanent increase in the financed share has a locally unique converging competitive-equilibrium transition. Initial mean fertility rises and the final population is larger. The original initial old keep their pre-reform financial assets and housing titles.

The construction below has an owner share of 11/21, so almost half the households rent. It proves local existence around an interior mixed economy, not just continuity from a zero-renter limit. The neighborhood and the permitted reform size are not given numerical lower bounds. It does not prove global convergence, arbitrary changes of binding constraints, or a general fertility sign. Its finite small reforms eventually oscillate around the new steady state: cumulative fertility can rise without fertility increasing at every date. The short direct-allocation welfare result remains separate.

2 Original choices and the maintained branch

Keep the original flow utilities and value functions. Write $x = c - \chi n$, $s = h - \kappa n$, and $u_t = qr_t$ only to shorten the proof. The primitives and the fertility weight ϑ are constant after the reform. There is no capital-gains tax. Define

$$\rho_O = 1 + \beta(1 + \gamma + \omega_B), \quad \rho_R = 1 + \beta(1 + \omega_B), \quad a = h_R^{\max}.$$

We maintain the following uniformly strict conditional branches. Young owners are at their down-payment limit, below their physical owner cap, and save positively. Young and old renters are at their rental caps and young renters save positively. Old owners are interior, with slack retention and estate bounds. Adult goods, adult space, fertility and estates are positive. Every conditional young owned size exceeds a . Conditional choices are evaluated in both tenures even when one is selected less often.

Let F be the fixed entrant distribution of income and wealth, with compact support and uniform margins for these inequalities. Additive ownership tastes have a finite positive logistic scale σ_ξ . Conditional policies are independent of that taste draw; ownership probabilities are not. The same probability must enter housing, births and subsequent old-age demand.

At date t , set $w_i = y_i + b_i + T_t + qT_{t+1}$. The original dated budgets and first-order conditions imply

$$\begin{aligned} h_i^O &= \frac{b_i}{(1 - \phi)P_t}, \quad h_i^R = a, \quad \rho_O x_i^O + \chi n_i^O = w_i - u_t h_i^O, \\ \rho_R x_i^R + \chi n_i^R &= w_i - u_t a - q u_{t+1} a, \quad \frac{\vartheta}{n_i^m} = \frac{\chi}{x_i^m} + \frac{\alpha \kappa}{h_i^m - \kappa n_i^m}. \end{aligned}$$

Old choices for these young households satisfy

$$c_{t+1,i}^{2,m} = \frac{\beta x_{t,i}^m}{q}, \quad h_{t+1,i}^{2,O} = \frac{\beta \gamma x_{t,i}^O}{q u_{t+1}}, \quad h_{t+1,i}^{2,R} = a, \quad e_{t+1,i}^m = \frac{\beta \omega_B x_{t,i}^m}{q^2}.$$

Saving and old net assets are reconstructed from the original young budget and mortgage repayment. In particular the estate is dated in next-period goods; the denominator above is q^2 .

3 Six variables suffice for aggregate equilibrium

Use $Z_t = (P_t, u_t, Y_t, O_t, M_t, R_t)$. The variables Y_t, O_t are cohort masses. The quantity M_t/u_t is total old-owner housing; M_t is not financial wealth. The variable R_t is total old-renter housing. For $t \geq 1$,

$$M_t = \frac{\beta\gamma}{q} Y_{t-1} \int \pi_{t-1}^O(i) x_{t-1,i}^O dF, \quad R_t = a Y_{t-1} \int (1 - \pi_{t-1}^O(i)) dF.$$

The user-cost identity and balanced tax rebate give

$$P_{t+1} = \frac{(1 + q\tau^p)P_t - u_t}{q}, \quad T_t = \frac{q\tau^p P_t \bar{H}}{Y_t + O_t}.$$

Given Z_t , take $v = (u_{t+1}, Y_{t+1})$ as two unknowns and set $T_{t+1} = q\tau^p P_{t+1} \bar{H} / (Y_{t+1} + Y_t)$. Evaluate the preceding original conditional choices and the logistic tenure probability. Solve

$$F_1(Z_t, v) = Y_t \int [\pi_i h_i^O + (1 - \pi_i) a] dF + M_t/u_t + R_t - \bar{H} = 0,$$

$$F_2(Z_t, v) = Y_{t+1} - \nu Y_t \int [\pi_i n_i^O + (1 - \pi_i) n_i^R] dF = 0.$$

The exact update is

$$Z_{t+1} = G(Z_t, v) = \left(P_{t+1}, u_{t+1}, Y_{t+1}, Y_t, \frac{\beta\gamma}{q} Y_t \int \pi_i x_i^O dF, a Y_t \int (1 - \pi_i) dF \right).$$

Where F_v is nonsingular, the implicit function theorem gives a smooth local map $Z_{t+1} = g(Z_t; \phi)$. This is a representation of the original equilibrium equations, including the future rental cost in tenure choice. It is not an assumed adjustment rule.

3.1 A simple sufficient condition for the forward solve

For homogeneous income and wealth, suppose also that each old owner occupies at least a . These conditions imply nonsingularity of F_v even with a positive property tax; neither a small-tax nor a small-child-cost condition is needed for this particular step.

To see this, housing clearing fixes the owner probability at the given state:

$$\pi = \frac{\bar{H} - M/u - R - Ya}{Y(h^O - a)} \in (0, 1).$$

Write $z = u_{t+1}$ and let w vary with the future rebate. The envelope derivatives of $\Delta = W^O - W^R$ are

$$\Delta_z = -\frac{\beta\gamma}{z} + \frac{qa}{x_R} < 0, \quad \Delta_w = \frac{1}{x_O} - \frac{1}{x_R}.$$

The strict inequality is exactly the old renter's cap condition. Holding π fixed therefore implies

$$\frac{dz}{dw} = \frac{1/x_O - 1/x_R}{\beta\gamma/z - qa/x_R}.$$

At fixed housing, fertility is increasing in resources when $\chi > 0$: $n_{m,w} > 0$. The renter's dependence on z is $n_{R,z} = -qa n_{R,w}$; the owner's is zero at fixed w . Hence along the fixed-tenure-probability equation,

$$\frac{d\bar{n}}{dw} = \pi n_{O,w} + (1 - \pi) n_{R,w} \frac{\beta\gamma/z - qa/x_O}{\beta\gamma/z - qa/x_R} \geq 0.$$

The numerator in the fraction is nonnegative precisely when $h^{2,O} \geq a$. Meanwhile w is nonincreasing in Y_{t+1} because the future tax revenue is rebated to $Y_{t+1} + Y_t$ households. After solving housing for z , the derivative of the birth-equation residual with respect to Y_{t+1} is $1 - \nu Y_t (d\bar{n}/dw)(dw/dY_{t+1}) \geq 1$. This proves local nonsingularity. At $\chi = 0$ the resource derivatives of fertility vanish and the same conclusion follows. This argument does not assert existence at distant states or global continuation across a branch change.

4 The actual initial old

Keep the complete pre-reform old distribution G_0 as exogenous data. Let B_0 be its owner mass, A_0 its aggregate owner net financial assets after the original mortgage repayment, and H_0 its aggregate inherited owner title. The latter is inherited purchased housing, not housing retained after the old reoptimize. Let $L = \gamma/(1 + \gamma + \omega_B)$.

The four initial restrictions are

$$Y = Y_0, \quad O = O_0, \quad R = a(O_0 - B_0), \quad M = L[A_0 + PH_0 + T(P, Y_0, O_0)B_0].$$

The initial owner coefficient varies with the surprising price and rebate; fixing it would wrongly suppress revaluation of existing title. These four aggregates suffice for date-zero housing clearing. The full G_0 is still needed to reconstruct and check each old household's consumption and estate. Uniform slack at the initial equilibrium preserves those inequalities under a small reform.

5 A local nonlinear transition theorem

Suppose the local map and boundary operator are at least C^2 , Z^* is a stationary equilibrium on this branch, F_v is nonsingular, and $J = g_Z(Z^*; \phi)$ has four eigenvalues strictly inside and two strictly outside the unit circle, counted with multiplicity. Let B be the derivative of the four initial restrictions with respect to Z_0 . Suppose B restricted to the four-dimensional stable generalized eigenspace is invertible. Then sufficiently small permanent changes of ϕ , starting from the original stationary population and claims, have a locally unique equilibrium sequence converging exponentially to the nearby stationary state.

Proof. Since 1 is not an eigenvalue, $(I - J)$ is invertible and the stationary state is a smooth function of the reform. Center the sequence on this new state. Split the linear recurrence into its stable and unstable generalized eigenspaces. Propagate the stable part forward and obtain its initial value from B ; solve the unstable part backward using its inverse and the requirement that the sequence converge. Choose a weight $\eta < 1$ above all stable moduli and above the inverse unstable moduli. The resulting operator has a bounded inverse on sequences with norm $\sup_t \eta^{-t} \|v_t\|$. The nonlinear residual and boundary conditions are smooth on a uniform neighborhood, so the sequence implicit function theorem applies. A zero stable eigenvalue is harmless: only the unstable block is inverted. Uniform household margins and differentiation under the fixed compact-support integral reconstruct the full original equilibrium. This proves uniqueness among nearby convergent sequences, not global uniqueness of equilibrium. $\square$

6 An exact mixed equilibrium

All the following primitive values are rational except the ownership taste location, which is specified by the exact value functions below:

$$\begin{aligned} q &= \frac{1}{2}, \quad \phi = \frac{4}{5}, \quad b = \frac{1}{5}, \quad \alpha = \beta = \omega_B = \frac{2}{5}, \quad \gamma = \frac{3}{10}, \\ \chi &= \frac{3}{20}, \quad \kappa = \frac{1}{2}, \quad \vartheta = \frac{141}{400}, \quad \nu = 2, \quad a = \frac{1}{4}, \quad h_O^{\max} = 2, \quad \sigma_\xi = 1, \\ \tau^p &= \frac{467}{9250}, \quad y = \frac{3521349281}{1677802000}, \quad \bar{H} = \frac{68104}{68019}. \end{aligned}$$

Set the logistic taste location to $\bar{\xi} = \sigma_\xi \log(\pi/(1-\pi)) - W^O + W^R$ at the allocations below, where $\pi = 11/21$. This is a primitive specification of a finite taste distribution, not an equilibrium-dependent policy rule. Hold it fixed during the credit reform. This construction is an analytical example, not a calibration.

There is an exact steady state with

$$\begin{aligned} P &= Y = O = 1, \quad u = \frac{9717}{18500}, \quad T = \frac{3975571}{314587875}, \quad \pi = \frac{11}{21}, \\ (x_O, h_O, n_O) &= (1, 1, \frac{3}{4}), \quad (x_R, h_R, n_R) = (\frac{99}{74}, \frac{1}{4}, \frac{9}{40}). \end{aligned}$$

Mean fertility is exactly 1/2. Old owner housing is $1480/3239 > a$, old owner consumption is $4/5$, and its estate is $16/25$. The strict original constraints are checked in the companion receipt. In particular the young owner's marginal value of space is $.64 > u$, whereas the interior old owner's equals u . Thus this same equilibrium also satisfies the short housing misallocation proposition.

6.1 Exact differentiation

Differentiate the two conditional resource and fertility equations. For $A_n = \vartheta/n^2 + \alpha\kappa^2/s^2$,

$$\begin{pmatrix} \rho_m & \chi \\ -\chi/x_m^2 & A_n \end{pmatrix} \begin{pmatrix} dx_m \\ dn_m \end{pmatrix} = \begin{pmatrix} dw - h_m du - u dh_m - \mathbf{1}_{m=R} qa du_{t+1} \\ \alpha\kappa dh_m/s_m^2 \end{pmatrix}.$$

The owner derivative is $dh_O = h_O d\phi/(1-\phi) - h_O dP/P$ and $dh_R = 0$. Envelope derivatives give

$$dW_O = \frac{dw - h_O du}{x_O} + (\alpha/s_O - u/x_O)dh_O - \frac{\beta\gamma}{u_{t+1}} du_{t+1}, \quad dW_R = \frac{dw - a du - qa du_{t+1}}{x_R},$$

and $d\pi = \pi(1-\pi)(dW_O - dW_R)/\sigma_\xi$. All coefficients at the specified equilibrium are rational. Therefore

$$J = G_Z - G_v F_v^{-1} F_Z, \quad Q = g_\phi = G_\phi - G_v F_v^{-1} F_\phi$$

are exact rational matrices. No numerical differentiation is used in their certificate. Separate complex-step differentiation of the original household helper checks the derivation.

6.2 Roots, initial conditions and signs

The characteristic polynomial is $zp_5(z)$ up to a positive constant, with

$$\begin{aligned} p_5(z) &= 11624093714366716003479040894500z^5 \\ &\quad + 258366763596378495144999871594252z^4 \\ &\quad - 1176628664469803650647365064033229z^3 \\ &\quad + 408376917290861767134959387855371z^2 \\ &\quad - 199454780254172995433079109028654z \\ &\quad + 694085946210757308455013341344. \end{aligned}$$

An exact Sturm count gives three real roots. Rational isolating intervals place them respectively near -26.1500318 , $.003504813$, and 3.5930940 . Vieta's formulas put the remaining pair's product near $.18132116$ and sum near $.32660224$. Rational inequalities prove that the pair is nonreal with modulus in $(.42, .44)$. There are exactly four stable roots including zero, and two unstable roots.

To check the boundary without rounded complex eigenvectors, let p_6 be the monic characteristic polynomial. A polynomial row $\ell(z)$ constructed from J satisfies exactly

$$\ell(z)(J - zI) = -p_6(z)e_P^\top.$$

It is nonzero at each unstable root. Those two left eigenvectors annihilate the stable generalized space. Rational interval Gaussian elimination proves that the determinant of the matrix whose rows are B , $\ell(r_-)$ and $\ell(r_+)$ excludes zero. After the normalization specified in the accompanying verification file, its value is near $.2580033$. This verifies the theorem's boundary condition.

The stationary derivative $Z_\phi^* = (I - J)^{-1}Q$ has

$$\frac{dY^*}{d\phi} = \frac{38021133978224611827644635285}{35866089542871200498536115024} > 0.$$

For the initial derivative dZ_0 , solve $BdZ_0 = 0$ and $\ell(r_\pm)dZ_0 = \ell(r_\pm)Z_\phi^*$. Then $dZ_1 = JdZ_0 + Q$. A rational enclosure proves

$$.0288 < \frac{d\bar{n}_0}{d\phi} < .0289, \quad 1.0600 < \frac{dY^*}{d\phi} < 1.0602.$$

The full exact enclosures are stored with the accompanying verification file receipt; the decimal bounds here are outward bounds for readability. Smoothness gives both signs for sufficiently small finite positive reforms. Strict margins, root counts, boundary transversality and strict response signs also persist in an open neighborhood of the primitives. Small nondegenerate compact-support perturbations of entrant income and wealth are included, retaining the uniform conditional branches. This is not a certified rectangular parameter box or a claim about arbitrary heterogeneity.

7 What happens after the initial fertility rise

The constructed transition need not be monotone. In fact every sufficiently small positive reform eventually places fertility on both sides of replacement, infinitely often; population similarly crosses its new steady level infinitely often. The crossings decay exponentially.

Here is a finite-reform argument. At the baseline let $v = dZ_0 - Z_\phi^*$, $w_1 = Jv$, and $w_2 = J^2v$. If the young-cohort response had no component from the complex pair, then $e_Y^\top w_2 - r_s e_Y^\top w_1 = 0$, where r_s is the small real root; the zero mode has disappeared after one step. The rational interval check instead places this expression near $-.1346566$, strictly below zero. Thus the complex modes occur in the young-population response.

For a small reform δ , write the deviations from its new steady state in smooth local stable coordinates. Let $\lambda(\delta)$ be one of the complex roots. Uniformly, $|\lambda| \in (.42, .44)$ and the other two stable roots have modulus below $.01$.

Uniform tail lemma. The stable graph and its restricted map may be chosen C^2 jointly in the state and δ . At zero state its nonlinear remainder and its state derivative are zero. On a sufficiently small common neighborhood Taylor's theorem therefore bounds the remainder by $C\|v\|^2$, its state

derivative by $C\|v\|$, and its parameter derivative by $C\|v\|^2$. For the last bound one may use C^3 regularity; here the conditional formulas, implicit map and parameter-dependent stable graph can be taken at least C^3 . The fixed compact entrant support allows their first three derivatives to pass under the integral. Choose a stable-coordinate norm whose linear contraction is strictly below .5. Shrinking the neighborhood preserves a nonlinear contraction by $\eta = .5$. The initial stable coordinate is a C^2 function of δ and vanishes at zero, so iteration gives $\|v_t\| \leq C|\delta|\eta^t$. Differentiating that iteration, or using the weighted sequence IFT, gives $\|\partial_\delta v_t\| \leq C\eta^t$. Consequently its nonlinear forcing and derivative are bounded respectively by $C\delta^2\eta^{2t}$ and $C|\delta|\eta^{2t}$. Constants are uniform for all sufficiently small δ .

The parameter-dependent graph can also be obtained directly by the one-sided sequence IFT, prescribing the stable initial coordinate and leaving the unstable initial coordinate free. That construction requires invertibility only on the unstable space, so the zero stable root causes no problem here either. This supplies the regularity used in the lemma. The complex coordinate therefore satisfies

$$a_{t+1} = \lambda a_t + r_t, \quad |r_t| \leq C\delta^2\eta^{2t}.$$

Since $\eta^2 < |\lambda|$, the convergent sum

$$A(\delta) = a_0 + \sum_{s=0}^{\infty} \lambda^{-s-1} r_s$$

gives $a_t = A(\delta)\lambda^t + O(\delta^2\eta^{2t})$. The fast stable coordinates are $O(|\delta|(.01)^t + \delta^2\eta^{2t})$; unstable coordinates on the local stable graph are quadratic in the stable coordinates. Consequently

$$Y_t - Y^*(\delta) = 2 \operatorname{Re}[C_Y(\delta)\lambda(\delta)^t] + O(|\delta|(.01)^t + \delta^2\eta^{2t}).$$

The preceding nonzero linear signal implies $C'_Y(0) \neq 0$, so $C_Y(\delta) \neq 0$ for every sufficiently small nonzero δ . This differentiability follows from the lemma and differentiation of the uniform convergent sum. Differentiating λ^{-s-1} adds a factor proportional to $s+1$, which remains summable because $\eta^2/|\lambda| < 1$. The nonlinear summands sum to $O(\delta^2)$.

A nonzero real projection of a rotating nonreal eigenmode has infinitely many strictly positive and negative values bounded away from zero after division by its modulus. For a rational rotation this follows over a period; for an irrational rotation it follows from density on the circle. The remainder above is smaller along both subsequences. Cohort population therefore crosses its limiting level infinitely often. Total household population has the same conclusion, since its leading coefficient is multiplied by $1 + 1/\lambda \neq 0$. Finally,

$$\bar{n}_t - \frac{1}{\nu} = \frac{Y_{t+1} - Y_t}{\nu Y_t},$$

whose leading complex coefficient is multiplied by $(\lambda - 1)/(\nu Y^*) \neq 0$. Fertility too approaches replacement from both sides. This result is compatible with higher initial fertility and a larger final population. It makes no all-date welfare claim.

8 A family of mixed economies

The same stationary allocation can be supported at different taste scales. Keep the listed primitives and stationary bundles, let $\sigma = \sigma_\xi > 0$, and specify $\tilde{\xi}(\sigma) = \sigma \log(11/10) - W^O + W^R$. This is a one-parameter family of primitive taste distributions with the same stationary allocation. Within

each economy, both taste parameters are fixed during the credit reform. Comparing different σ here is not a policy experiment.

Proposition. Every member with $1 \leq \sigma \leq 4$ has a locally unique converging transition after a sufficiently small permanent credit relaxation. Initial fertility and final population rise. The result persists under small changes in the other primitives and small entrant heterogeneity with uniform branch margins. The constructed family's owner share is always 11/21; nearby economies need not have that exact share. The oscillation theorem in Section 7 is established near $\sigma = 1$, not on the entire interval. For every $\sigma > 0$, the stationary young-cohort derivative is positive and each tenure's stationary lifetime-value derivative is negative.

The stability part is algebraic for the entire positive half-line. Exact differentiation gives $J(\sigma) = J(0) + \sigma \mathbf{a} \mathbf{b}^\top$ for two constant vectors $\mathbf{a}, \mathbf{b}$, and $Q(\sigma)$ affine in σ . Here $J(0)$ is the algebraic limit of the matrices; a degenerate taste distribution is not assumed in the economic theorem. The nonzero-root polynomial is affine in σ :

$$p_{5,\sigma}(z) = p_{5,1}(z) + (\sigma - 1)p_1(z),$$

where $p_{5,1}$ is the integer polynomial in Section 6 and

$$\begin{aligned} p_1(z) = & 209842616525591903123960392500000z^4 \\ & - 1024713381846718023989705716419600z^3 \\ & + 426827306877372594433334439987600z^2 \\ & - 162673088007827714095534626813600z \\ & + 495225810071736959616928665600. \end{aligned}$$

Apply $z = (1 + w)/(1 - w)$ to this quintic. The first column of the Routh array for $(1 - w)^5 p_{5,\sigma}((1 + w)/(1 - w))$ has signs

$$-, \quad -, \quad -, \quad +, \quad +, \quad -.$$

Each sign follows because the corresponding rational function of σ has numerator coefficients all of the displayed sign and denominator coefficients all positive. The exact coefficients are provided with the reproducible proof. No row vanishes. Thus exactly two roots have positive real part in w , and three have negative real part. There are two roots outside the unit circle in z and three inside, in addition to the zero root of J . The quintic's discriminant is a degree-eight polynomial in σ with all coefficients negative. Together with the three-real-root count at one point, this proves three distinct real roots and one nonreal pair for every $\sigma > 0$. The signs at $z = -1, 1$ and at infinity identify one real root below -1 and one above 1 .

For the boundary and initial fertility sign, the rational interval check covers the entire closed interval $[1, 4]$ with 913 adjacent subintervals. At each subinterval's endpoints, exact root isolation brackets the two unstable real roots. A bracket whose polynomial signs agree at both parameter endpoints brackets the root for every intermediate parameter, since the polynomial is affine in σ . This is an interval proof, not a grid of unchecked intermediate economies.

The rank-one representation provides a polynomial left eigenvector independent of σ . For $A = J(0)$, take $\ell(z) = \mathbf{b}^\top \text{adj}(zI - A)$. The exact identity

$$\ell(z)(J(\sigma) - zI) = -p_{6,\sigma}(z)\mathbf{b}^\top$$

follows from the determinant lemma and is also checked coefficient by coefficient. Interval elimination then establishes a nonzero initial-boundary determinant and a strictly positive initial fertility

derivative throughout each parameter interval. Adjacent endpoints and full coverage are verified exactly. The family receipt gives uniform positive lower bounds; the allowed finite reform size is still not numerically quantified.

There is a simpler direct calculation for the stationary comparison. The common denominator is positive:

$$D(\sigma) = 29694400415045569770000\sigma + 7922531612096952274579 > 0.$$

Exact differentiation yields

$$\frac{dY^*}{d\phi} = \frac{55(625392361132072652215364400\sigma + 65900983926556653741810787)}{953456D(\sigma)} > 0,$$

$$\frac{dW^{O*}}{d\phi} = -\frac{33(568192684411159099406576\sigma + 7091548916248680863487)}{224D(\sigma)} < 0,$$

$$\frac{dW^{R*}}{d\phi} = -\frac{83313(49680061080395802000\sigma + 2808938751889938767)}{224D(\sigma)} < 0.$$

These signs hold for every positive σ . At $\sigma = 1$, the value derivatives are approximately -2.25302 for owners and $-.518979$ for renters. Since both conditional values fall and the additive ownership taste is fixed, the maximum of the two values falls for every fixed taste draw. This compares the same entrant in two steady states. It does not rank different populations socially or describe all transitional cohorts. In particular, the initial old may benefit from the rise in the value of their inherited titles. The competitive credit reform does not itself provide the compensation used in the housing-allocation proposition.

The accompanying evidence index gives the exact calculations and regeneration command. The checks include rational root and sign bounds, complete interval coverage, and independent comparisons with the original dated household equations. Finite numerical paths check those equations; local convergence follows from the proof above. The earlier all-owner transition and its primitive conditions remain in the longer assessment.